Mathematical Pathology and Geometric Universality in Diffusion Limited Aggregation

Candidate Number: [Your Number]

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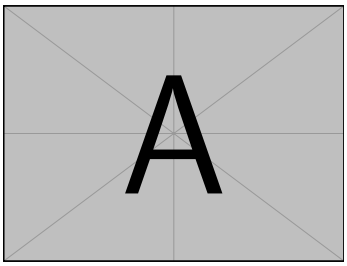

A dissertation submitted for the degree of
Mathematics Part C

Abstract

This dissertation investigates the universality class of Diffusion Limited Aggregation (DLA) through the lens of conformal mapping and harmonic measure theory. We compare on-lattice, off-lattice, and hybrid "snapping" algorithms to determine if microscopic lattice anisotropy survives in the macroscopic scaling limit. Particular attention is paid to the dimension of the harmonic measure and the dynamics of the active growth zone. [Placeholder: Finish this summary with your specific results regarding the Hybrid model's classification, totaling approx. 300 words.]

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Chapter 1

Introduction

1.1 DLA Overview & Applications

Diffusion Limited Aggregation (DLA) is a process where particles undergo a random walk and cluster together to form fractals. This section will define the basic growth rule and discuss physical applications such as electrodeposition, dielectric breakdown, and Hele-Shaw flow.

1.2 History of Results

We trace the history of DLA from the original Witten-Sander paper (1981) through the early numerical estimates of fractal dimension. This section outlines the progression from purely numerical observations to the development of rigorous mathematical bounds by Kesten and others.

1.3 Key Mathematical Concepts

This section introduces the core machinery required to analyze DLA: Brownian motion as the continuum limit of random walks and its relationship to the Laplace equation ($\nabla^2\phi = 0$). We define the Harmonic Measure, explaining its crucial role in determining the probability of a particle sticking to a specific site on the cluster boundary.

1.4 The “Universality” Debate

Here we frame the central problem of the dissertation: does the underlying grid geometry (lattice vs. off-lattice) fundamentally change the fractal dimension? We introduce the concept of "universality classes" and the controversy surrounding whether hybrid models belong to the continuum or lattice class.

Chapter 2

Algorithms

2.1 Conformal Maps & Harmonic Measure Theory

We discuss the Hastings-Levitov algorithm and how conformal mapping can theoretically generate DLA clusters without random walks. This section connects the physical growth probability to the derivative of the mapping function from the unit circle to the cluster exterior.

2.2 On-Lattice Algorithm

This section details the standard square-lattice implementation and the issue of grid bias. We explain Loh's algorithm or similar "bias-free" modifications that attempt to suppress the anisotropy inherent in the 4-neighbor connectivity.

2.3 Off-Lattice Algorithm

We describe the standard off-lattice Monte Carlo simulation where particles move in continuous space ($\mathbb{R}^2$). This includes optimization techniques like allowing large jump steps when the particle is far from the cluster (using Green's functions).

2.4 Hybrid Model (Snapping Logic)

This section defines the specific "Hybrid" model under investigation. We explain the "snapping" logic where particles move off-lattice but stick to the nearest grid point upon collision, effectively mixing continuum transport with discrete aggregation.

Chapter 3

Analysis

3.1 Global Scaling

We present results on the mass-radius relationship $M(R) \sim R^D$. This section calculates the fractal dimension D for the different algorithms generated in Chapter 2 and compares them to the accepted value of $D \approx 1.71$.

3.2 Anisotropy (Angular Growth Rate)

Using the method proposed by Grebenkov or similar angular harmonic analysis, we quantify the anisotropy of the clusters. We test whether the clusters exhibit 4-fold symmetry (indicating lattice pathology) or isotropic growth.

3.3 Active Zone Dynamics

This section analyzes the "active zone"—the set of perimeter sites with non-negligible harmonic measure. We distinguish between the "frozen interior" of the cluster and the active tips, discussing how the depth of the fjords affects particle penetration.

Chapter 4

Discussion

4.1 Synthesis of Results

We bring together the scaling and anisotropy data to draw conclusions about the Hybrid model. This section argues whether the "snapping" mechanism introduces enough noise to wash out lattice effects or if the underlying grid still dictates the asymptotic shape.

4.2 Final Classification

We provide a definitive classification of the Hybrid model within the universality landscape. We conclude whether DLA is truly geometrically universal or if mathematical pathologies persist in semi-discrete models.

Bibliography

T. A. Witten and L. M. Sander. Diffusion-limited aggregation, a kinetic critical phenomenon. *Physical Review Letters*, 47(19):1400, 1981.

Appendix A

Code Snippets and Derivations

This appendix contains the Python/C++ implementation of the random walk loop and the derivation of the hitting probability for a circle in 2D Brownian motion.

Appendix B

Reference: LaTeX Structure Examples

This section provides templates for common mathematical structures used in this dissertation.

B.1 Aligned Equations

Below is a multi-line derivation showing the alignment of the equals sign using the `align` environment.

$$\begin{aligned} \nabla^2\phi(r,\theta) &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} && \text{(B.1)} \\ &= 0 \quad \text{(Laplace Equation)} \\ \implies \phi(r,\theta) &= \sum_{n=0}^{\infty} r^n(A_n\cos n\theta + B_n\sin n\theta). && \text{(B.2)} \end{aligned}$$

B.2 Floats

B.2.1 Single Figure

Figure B.1 shows a standard inclusion of a graphic.

B.2.2 Side-by-Side Figures (Subfloats)

Figure B.2 demonstrates how to place two plots next to each other for comparison.

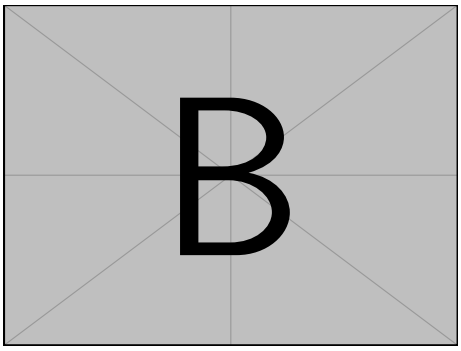
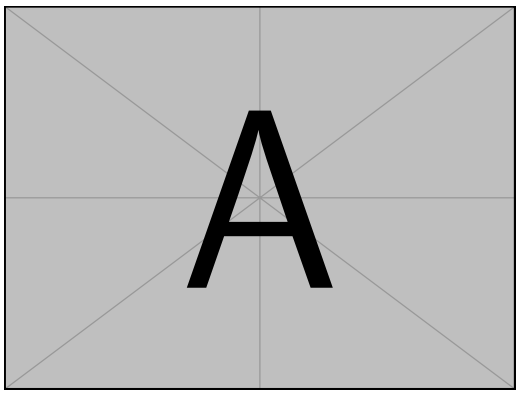
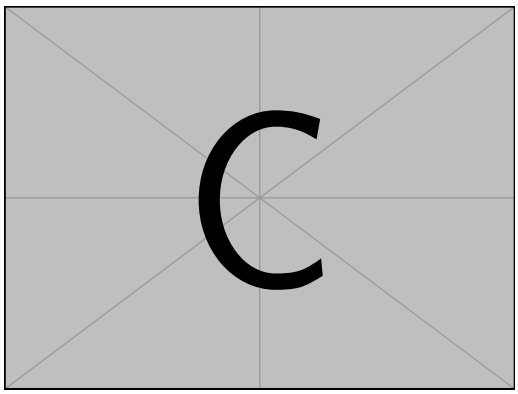


Figure B.1: A realization of 2D Brownian motion.



(a) On-Lattice DLA



(b) Off-Lattice DLA

Figure B.2: Comparison of cluster morphology between lattice and continuum models.

B.3 Algorithms

Algorithm 1 shows a basic random walk loop using the `algorithm2e` package.

Algorithm 1: Basic Random Walk Simulation

Result: Final position of particle

```
Initialize particle at r_start;
while particle not stuck AND particle within bounds do
    Generate random angle θ ∈ [0, 2π);
    x ← x + δ cos θ;
    y ← y + δ sin θ;
    if distance to cluster < ϵ then
        Stick particle;
        break;
    end
end
```

B.4 Theorems

Standard environments for formal mathematical statements.

Definition B.1 (Harmonic Measure)**.** *The harmonic measure $\omega(z, E, \Omega)$ is the probability that a Brownian motion starting at $z \in \Omega$ first hits the boundary $\partial\Omega$ in the set $E \subset \partial\Omega$.*

Theorem B.1 (Kakutani's Theorem)**.** *The solution to the Dirichlet problem for the Laplace equation can be expressed stochastically in terms of the hitting distribution of Brownian motion.*

Proof. The proof follows from the martingale property of harmonic functions composed with Brownian motion. See Witten and Sander [1981] for details. □

B.5 References

This is an example of a citation. Witten and Sander introduced the model in 1981 Witten and Sander [1981].